Solution Design Document (SDD)

Actual vs Target Financial Performance Analysis Dashboard

Technology: Microsoft Power BI
Data Source: Microsoft Excel
Architecture: Dimensional / Star Schema
Primary Analytical Tool: DAX
Data Transformation: Power Query

1. Solution Overview

The proposed solution is an interactive Power BI reporting application designed to analyze financial performance against predefined targets.

The solution follows a structured Business Intelligence workflow:

Data Source → Data Transformation → Data Model → DAX Measures → Visualization → Business Insights

The architecture is designed to provide accurate calculations, interactive filtering, and scalable reporting.

2. High-Level Architecture

```
    Excel Dataset
       │
       ▼
    Power Query
       │
  Data Transformation
       │
       ▼
    Data Model
       │
       ▼
    DAX Measures
       │
```

▼

Power BI Visuals

│

▼

Interactive Dashboard

│

▼

End Users

3. Source Data Design

The source workbook contains four tables.

3.1 Actual

Contains actual financial performance values for each salesperson across monthly reporting periods.

Key attribute:

- Sales Person

Measures:

- Monthly Actual Value

3.2 Targets

Contains target financial values assigned to each salesperson across monthly reporting periods.

Key attribute:

- Sales Person

Measures:

- Monthly Target Value

3.3 Calendar

The Calendar table provides the date dimension required for time-based reporting.

It is used to support:

- Monthly analysis

- Trend analysis
- Date filtering
- Time-based calculations

3.4 dimPeople

The dimPeople table stores salesperson-related attributes.

Fields include:

- Sales Person
- Team
- Picture

The table acts as a dimension for salesperson and team analysis.

4. Data Transformation Design

The source Actual and Targets tables contain monthly values as separate columns.

For analytical modeling, these monthly columns are transformed using **Power Query Unpivot**.

Source Structure

Sales Person | Jan | Feb | Mar | Apr

Analytical Structure

Sales Person | Date | Value

This transformation provides a normalized structure suitable for Power BI modeling and time-series analysis.

Transformation Steps

1. Import Excel workbook.
2. Select Actual and Targets tables.
3. Validate column names.
4. Unpivot monthly columns.
5. Rename attribute/value columns.
6. Convert date columns to Date data type.
7. Convert financial values to numeric data type.
8. Remove unnecessary fields.

9. Validate missing and duplicate records.

10. Load transformed data into the Power BI model.

5. Data Model Design

The solution follows a dimensional modeling approach.

```
        Calendar
          |
          |
          ▼
        Actual
          |
          |
     dimPeople
       |    |
       |    |
       ▼    ▼
     Actual  Targets
          ▲
          |
        Calendar
```

Dimension Tables

- Calendar
- dimPeople

Fact Tables

- Actual
- Targets

The dimension tables provide filtering and descriptive context, while the fact tables contain measurable financial values.

6. Relationships

Salesperson Relationship

dimPeople[Sales Person]

│

▼

Actual[Sales Person]

dimPeople[Sales Person]

│

▼

Targets[Sales Person]

Date Relationship

Calendar[Date]

│

▼

Actual[Date]

Calendar[Date]

│

▼

Targets[Date]

The recommended relationship design is **one-to-many**, with dimension tables filtering the corresponding fact tables.

7. DAX Measure Design

The dashboard uses DAX measures for dynamic financial calculations.

Total Actual Value

TOTAL ACTUAL VALUE =

SUM(Actual[Value])

Total Target Value

TOTAL TARGET VALUE =

SUM(Targets[Value])

Variance

VARIANCE =

[TOTAL ACTUAL VALUE] - [TOTAL TARGET VALUE]

Variance Percentage

VARIANCE % =

DIVIDE(

  [VARIANCE],

  [TOTAL TARGET VALUE],

  0

)

The Variance % measure is formatted as a percentage.

8. Visualization Design

8.1 KPI Cards

The dashboard displays key financial metrics:

- Total Actual Value
- Total Target Value
- Variance
- Variance %

These provide an immediate overview of financial performance.

8.2 Actual vs Target KPI

The KPI visual compares actual performance against the target and provides a time-based performance trend.

8.3 Salesperson Performance Table

The detailed table provides:

- Salesperson
- Team
- Actual Value

- Target Value
- Variance
- Variance %
- Performance trend

This enables users to compare individual performance.

8.4 Variance Trend

The trend visualization displays how the Actual vs Target variance changes over time.

It helps users identify:

- Positive performance periods.
- Negative performance periods.
- Improving trends.
- Declining trends.

9. Interactive Filtering

The dashboard provides interactive slicers for:

Sales Person

Allows users to analyze the performance of a specific salesperson.

Team

Allows users to analyze performance at the team level.

When a filter is applied, the relevant KPIs and visualizations update dynamically.

10. Functional Requirements

ID	Requirement
FR-01	Display total actual financial value
FR-02	Display total target financial value
FR-03	Calculate financial variance
FR-04	Calculate variance percentage
FR-05	Provide salesperson filtering
FR-06	Provide team filtering

ID	Requirement
FR-07	Display monthly performance trends
FR-08	Display salesperson-level performance
FR-09	Dynamically update visuals based on filters
FR-10	Support Actual vs Target comparison

11. Non-Functional Requirements

Performance

Dashboard visuals should respond efficiently to user interactions and filters.

Accuracy

All financial metrics and calculated measures should accurately reflect the underlying source data.

Usability

The dashboard should provide a clear and intuitive interface for business users.

Maintainability

Power Query transformations and DAX measures should be organized to simplify future modifications.

Scalability

The data model should support additional reporting periods and business records without requiring major structural changes.

12. Validation and Testing

The following validations should be performed before deployment.

Test	Expected Result
Actual value validation	Matches source data
Target value validation	Matches source data
Variance validation	Actual – Target
Variance % validation	Variance ÷ Target
Salesperson filter	Displays selected salesperson

Test	Expected Result
Team filter	Displays selected team
Date filtering	Displays selected period
KPI validation	Updates correctly
Table validation	Displays correct salesperson metrics
Trend validation	Displays correct monthly trend

13. Security Considerations

If the solution is deployed within an organization, Power BI **Row-Level Security (RLS)** can be implemented.

For example:

Management

↓

All Teams

Team Manager

↓

Assigned Team

Salesperson

↓

Authorized Records

This can restrict access to sensitive financial information according to user roles.

14. Deployment Architecture

The completed Power BI report can be published to Power BI Service.

Power BI Desktop

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Power BI Service

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Workspace

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Published Report

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Authorized Users

The dataset can be configured for scheduled refresh when updated source data becomes available.

15. Future Enhancements

Potential future enhancements include:

- Target Achievement %
- Year-over-Year analysis
- Month-over-Month growth
- Top and Bottom Performer analysis
- Team ranking
- Performance forecasting
- Drill-through reports
- Automated data refresh
- Row-Level Security

- Mobile-optimized reporting
- Additional financial KPIs

16. Conclusion

The proposed solution provides a scalable and interactive approach to financial performance monitoring.

The combination of **Excel, Power Query, Power BI data modeling, DAX, and interactive visualizations** enables the transformation of raw Actual and Target data into a structured analytical reporting solution.

The architecture supports salesperson, team, and time-based analysis while providing management with a clear view of financial performance against targets.